

THAPAR TRADE

A Campus Marketplace for Student-to-Student Trading

Project Proposal for UCS503P

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1 Higher-order goal

The project aims to provide a structured and reliable campus marketplace for student-to-student exchange of used products at Thapar Institute of Engineering and Technology. Students commonly need to buy or sell textbooks, calculators, cycles, electronics, furniture, hostel appliances and other academic or personal items. These exchanges are currently facilitated largely through informal channels such as WhatsApp groups, ION groups, personal contacts and word of mouth.

2 Problem Statement

Although informal channels enable communication, they do not provide the functionality of a structured marketplace. Product information becomes distributed across multiple conversations, making relevant listings difficult to discover. Buyers cannot reliably search or filter products by category, price, condition or availability. Sellers may also fail to update or remove listings after an item has been sold, resulting in unnecessary communication and inaccurate availability information.

A further challenge is the reliability of marketplace listings. For example, a student may jokingly or misleadingly list a product belonging to another student as if it were their own, even though the actual product is not for sale. A general marketplace does not provide a campus-specific mechanism through which such listings can be reported and reviewed.

Key bottlenecks include:

- **Fragmented discovery:** Listings are distributed across multiple informal communication channels.
- **Weak searchability:** Buyers cannot consistently filter products by category, price, condition and availability.
- **Stale listings:** Sold products may remain visible because sellers do not update their posts.
- **Misleading listings:** Unauthorized, prank or false-ownership listings may be difficult to identify and resolve.
- **Lack of moderation workflow:** Existing channels do not provide a structured report-to-review process for disputed listings.

3 Proposed Solution

3.1 Overview

Build a campus-specific web marketplace with authenticated student accounts, structured product listings, search and filtering, availability management, reporting and administrative moderation. The platform will allow students to publish standardized listings and buyers to discover products through a searchable marketplace.

The proposed solution consists of the following components:

- **Student Marketplace:** Interface for browsing, searching, filtering and viewing product listings.
- **Authentication and User Management:** Associates marketplace activity with an authenticated student account.
- **Product Listing Module:** Captures title, category, description, price, condition and images, with validation before publication.
- **Listing Lifecycle Manager:** Maintains *Available*, *Reserved* and *Sold* states.
- **Reporting Mechanism:** Allows users to report misleading, unauthorized, inappropriate or falsely claimed listings.
- **Administrative Moderation:** Provides a dashboard for reviewing reports and temporarily restricting, removing or restoring listings.

3.2 Core workflow

The principal marketplace workflow is:

1. A student authenticates with the platform.
2. The seller creates a listing by entering standardized product information.
3. The system validates required fields before publication.
4. The listing is published with an *Available* status.
5. Buyers search and filter the marketplace to locate relevant products.
6. A buyer views product and seller information and contacts the seller.
7. The listing can transition to *Reserved* and subsequently *Sold*.

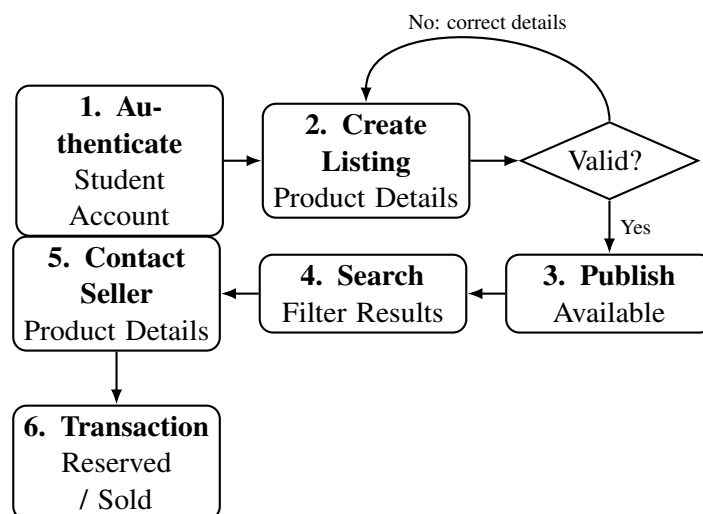


Figure 1: Thapar Trade Core Marketplace Workflow

3.3 Reporting and moderation workflow

The reporting mechanism is designed to address cases in which a student lists another student's product without authorization, including misleading or prank listings. A user can select "Report Listing" from the product page and choose a relevant reason such as product not for sale, actual ownership, false ownership claim, misleading information or other.

The report is stored in the database and becomes available to the administrator through the moderation dashboard. The administrator reviews the evidence and either restores the listing, removes it, or temporarily restricts it where appropriate.

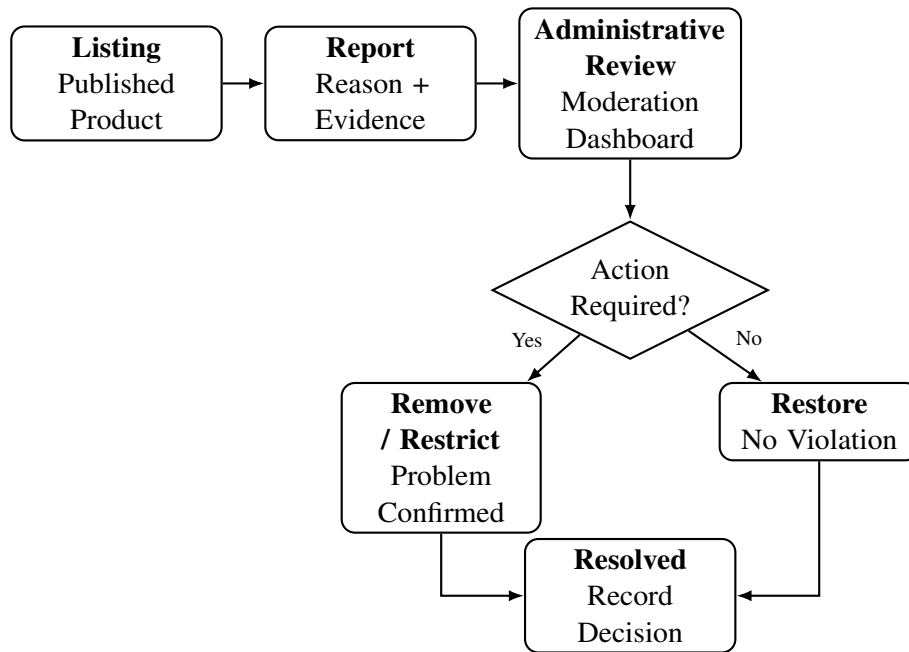


Figure 2: Reporting and Administrative Moderation Workflow

4 Solution Approach

This project focuses on building a reliable campus marketplace as an integrated software-engineering system rather than introducing complex research-driven algorithms. Development will proceed incrementally, beginning with the database schema and authentication layer, followed by listing creation and marketplace functionality. Search, filtering, status management, reporting and moderation will then be integrated with the core system.

4.1 System architecture

The proposed system uses a three-layer architecture consisting of a presentation layer, application layer and data layer. The separation supports modular development and keeps user-interface concerns, marketplace logic and persistent data management distinct.

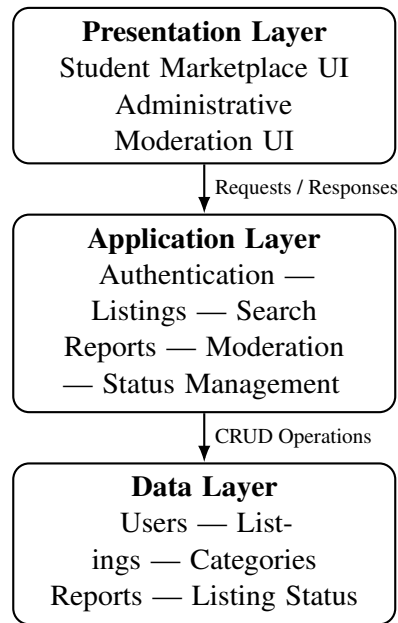


Figure 3: Thapar Trade 3-Layer Web Architecture

The presentation layer will provide responsive interfaces for students and administrators. The application layer will implement authentication, listing management, search, reporting and moderation logic. The data layer will maintain structured records for users, listings, categories and reports. Final implementation technologies will be selected according to team familiarity, development requirements and availability of free or academic-tier services.

4.2 Functional modules

The system will be implemented through the following modules:

- **Authentication and User Management:** The platform will associate marketplace activity with an authenticated student account.
- **Product Listing:** Sellers will provide standardized information including product title, category, description, price, condition and images. Required fields will be validated before publication.
- **Marketplace Search:** Students will search listings using keywords and refine results using category, price, condition and availability filters.
- **Listing Lifecycle:** Sellers will manage product availability using *Available*, *Reserved* and *Sold* states.
- **Product Details and Seller Information:** Buyers will view product information, listing condition, price and relevant seller information before contacting the seller.
- **Reporting Mechanism:** Users will report listings they believe are misleading, unauthorized, inappropriate or falsely claim ownership.
- **Administrative Moderation:** Administrators will review submitted reports and may temporarily restrict, remove or restore listings based on the review.

4.3 System execution and integration

The integrated system will operate through the workflow below. The validation stage prevents incomplete listings from reaching the public marketplace, while the listing-state stage maintains accurate availability.

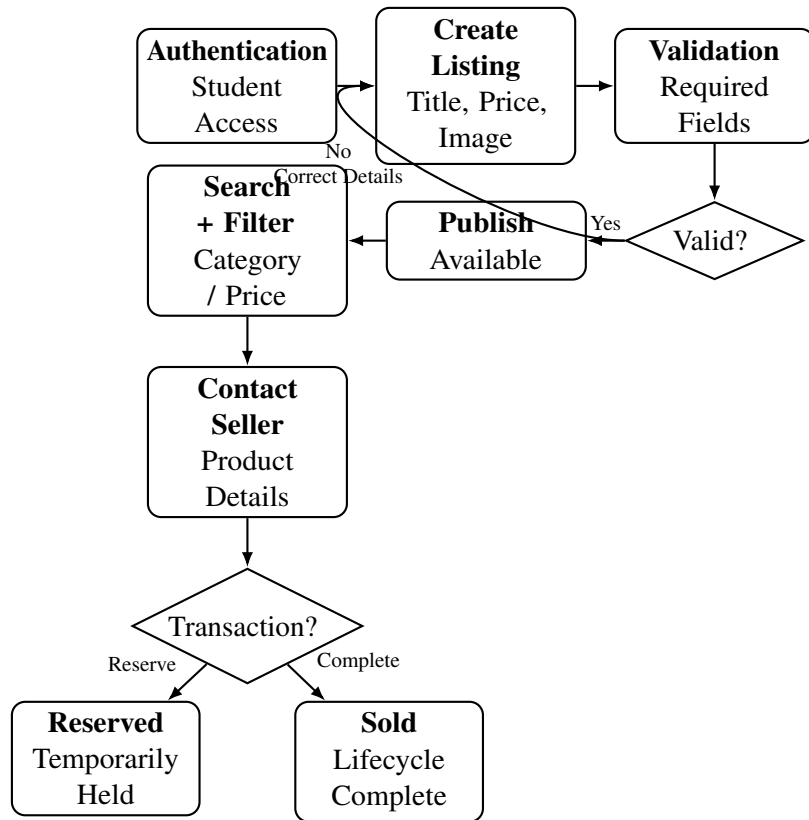


Figure 4: System Execution and Listing Lifecycle

5 Evaluation Criterion

Testing will be conducted at feature, integration and user-acceptance levels. Individual functions such as authentication, listing creation, search, status updates and reporting will first be tested independently. Integrated workflows will then be tested to ensure that data remains consistent between the user interface, backend and database.

A small group of approximately 10–20 students will participate in user-acceptance testing. Representative tasks will include:

- Locating a product using search and filters.
- Creating and publishing a product listing.
- Updating a listing from Available to Sold.
- Reporting a suspicious or unauthorized listing.
- Reviewing a reported listing through the administrative interface.

Evaluation will consider task completion, errors, time required and user feedback. The findings will be used to refine the interface and workflow before the final demonstration.

5.1 Testing workflow

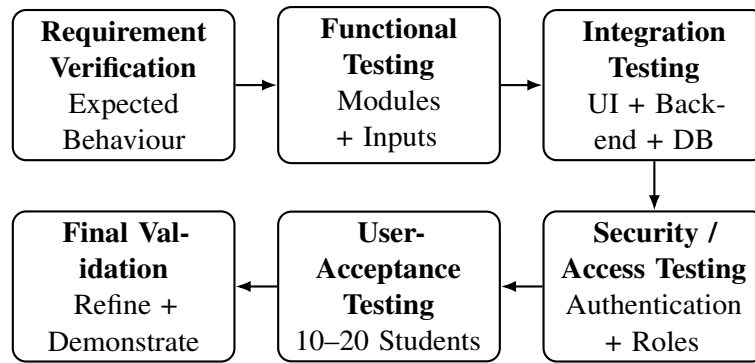


Figure 5: Testing and Validation Pipeline

6 Operational Constraints

The project will focus on a realistic semester-scale prototype. The system will not claim to automatically verify physical ownership of arbitrary products. Instead, it will provide authenticated student identities, seller declarations, reporting and administrative moderation.

Payment processing, delivery services, automated fraud detection, AI-based ownership detection and complex recommendation systems are outside the initial scope. Advanced features may be considered only after the core marketplace is stable.

7 Scalability and Availability

The architecture is intended to support future growth without changing the core marketplace workflow. Separating the presentation, application and data responsibilities allows individual modules to be improved independently. Free or academic-tier hosting and database services will be preferred during development, while the design can later be migrated to larger infrastructure if required.

The principal availability considerations are:

- Keep listing and search operations lightweight so normal student traffic remains responsive.
- Store listing and report records in structured persistent storage.
- Keep moderation operations separated from the normal student browsing workflow.
- Use modular interfaces so deployment infrastructure can be replaced without redesigning the application logic.

8 Project Scope and Deliverables

8.1 Initial deliverable

- Authenticated student marketplace interface.
- Product listing creation with structured fields and validation.
- Marketplace browsing, search and filtering.
- Available / Reserved / Sold listing lifecycle.

- Product details and seller information.
- Basic reporting and administrative moderation interface.

8.2 Subsequent deliverables

- Improved search and filtering behaviour.
- Refined moderation dashboard and reporting records.
- User-acceptance testing with student participants.
- Deployment of the integrated prototype and final documentation.

9 Risks and Mitigations

- **Misleading listings:** Provide reporting reasons and an administrative review workflow so disputed listings can be investigated.
- **Stale availability:** Use explicit Available, Reserved and Sold states and allow sellers to update listing status.
- **Incomplete listings:** Validate required fields before publication.
- **Integration defects:** Perform feature-level tests followed by end-to-end integration and user-acceptance testing.
- **Limited development resources:** Prioritize the complete core marketplace before considering advanced features.

10 Timeline

Weeks	Development Phase	Expected Deliverable
1–2	Requirements analysis and system design	Requirements, user flows, database design and UI prototype
3–4	Authentication and backend foundation	Working authentication and core backend services
5–7	Marketplace and listing module	Create, edit, browse and view product listings
8–9	Search and listing management	Search, filters and Available/Reserved/Sold workflow
10	Reporting and administrative moderation	Report submission, moderation dashboard and listing actions
11	Integration and user testing	Integrated prototype and student evaluation
12	Refinement and buffer	Final testing, documentation and presentation

Table 1: Proposed Semester Development Timeline

The final phase includes a buffer period for unexpected development, integration or testing delays.

11 Resources and Budget

The project will be developed by the three-member team. Work will be distributed across frontend development, backend and database development, interface design, testing, documentation and system integration. Team members will collaborate during integration and validation.

Required resources include student laptops, a code editor, Git/GitHub for version control, a database, web-development frameworks and an appropriate deployment environment. Product images will require suitable storage during development and testing.

Resource	Expected Cost
Development tools and code editor	No additional cost
Git/GitHub	No additional cost
Database services	Free/academic tier where available
Hosting/deployment	Free/academic tier where available
Testing devices	Existing student devices

Table 2: Resources and Estimated Project Cost

No specialized laboratory equipment or external funding is required for the proposed academic prototype. Free or academic-tier development and hosting services will be preferred where available.

12 Summary

This project proposes Thapar Trade, a deployable campus marketplace that converts informal student-to-student trading into a structured software workflow. The system emphasizes authenticated users, standardized listings, searchable products, explicit availability states and a report-to-moderation mechanism for misleading or unauthorized listings.

The project is intentionally scoped as a realistic semester-scale prototype, prioritizing a complete and reliable core marketplace before advanced functionality.